Kenya Cholera Early Warning System ⚠

(K-CEWS)

A Group 7 Presentation

The Team

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Introduction

Cholera is an acute diarrheal infection caused by the ingestion of food or water contaminated with the bacterium *Vibrio cholerae*.

The Clinical Reality: Known as a "disease of speed," it can kill a healthy adult within hours if untreated. While 80% of cases are treatable with **Oral Rehydration Salts (ORS)**, severe cases require intravenous fluids and antibiotics.

The Historical Context: Modern epidemiology began in **1854** when **Dr. John Snow** mapped a London outbreak to a contaminated water pump. Despite this, cholera remains endemic in Kenya since its 1971 entry.

The National Toll (Kenya): In the 2023-2024 cycle, Kenya reported over **12,000 cases** and **200+ deaths**. In many remote regions, the Case Fatality Rate (CFR) frequently exceeds the WHO 1% safety threshold.

Problem Statement

Current epidemiological surveillance in Kenya relies on the Integrated Disease Surveillance and Response (IDSR) framework. While robust for tracking, it suffers from three critical failures:

1. **The Lag Gap:** Systems trigger only after patients arrive at clinics, creating a 10–14 day Reactive Lag between environmental contamination and public health response.

2. **Resource Inefficiency:** Supplies are deployed after "Case Spikes," meaning resources often arrive *after* the peak of the transmission curve.

3. **Data Silos:** Environmental data (NASA), health data (WHO), and demographic data (KNBS) exist in isolation, preventing a holistic understanding of risk.

These ongoing failures thereafter lead to delayed cholera outbreak detection.

Objectives

General Objective

To develop an AI-driven predictive surveillance system (**K-CEWS**) that utilizes hydro-climatic and socio-demographic data to forecast cholera outbreak probabilities **14 days in advance** at the sub-county level in Kenya.

Specific Objectives

1. To integrate multi-source data streams from **NASA POWER** (Climate), **WHO** (Health), and **KNBS** (Census).
2. To implement **Feature Lagging (t-14 days)** to synchronize climate shocks with clinical onset.
3. To benchmark a multi-model pipeline optimized for **100% Recall (Sensitivity)**.
4. To deploy a high-availability **Streamlit Dashboard** for real-time decision support.

Data Understanding

The system integrates four isolated data streams into a continuous geographic time-series dataset spanning a 10-year period (2015-2025)

The final master geospatiotemporal dataset consists of;

- **27,324 rows**
- **17 engineered features**

Provides a robust foundation for high-sensitivity supervised learning using the **0.35 Surveillance Threshold**

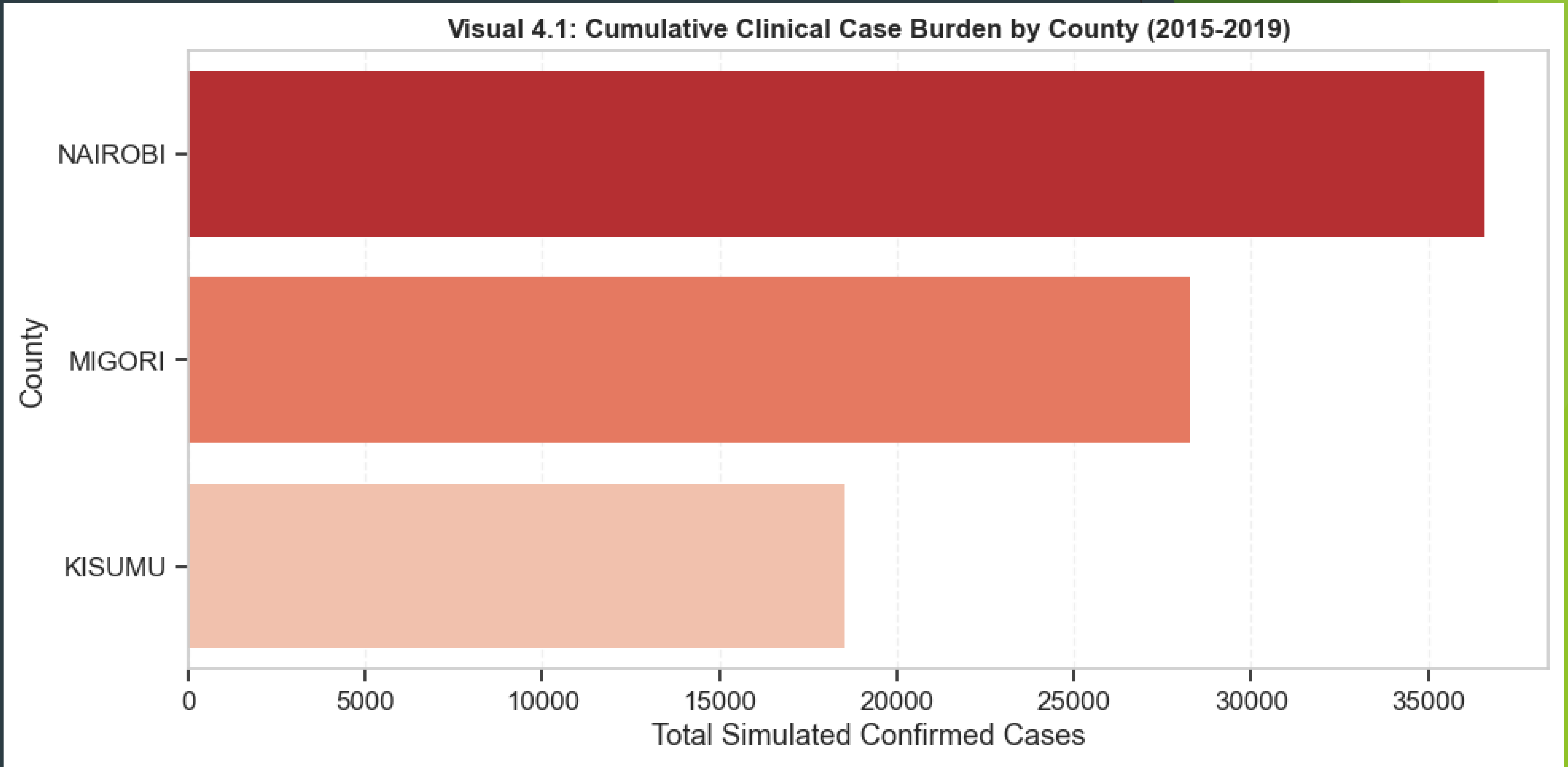
National Burden & Geospatial Hotspots

Key Finding:

Nairobi and Migori represent the primary "Reservoirs of Risk."

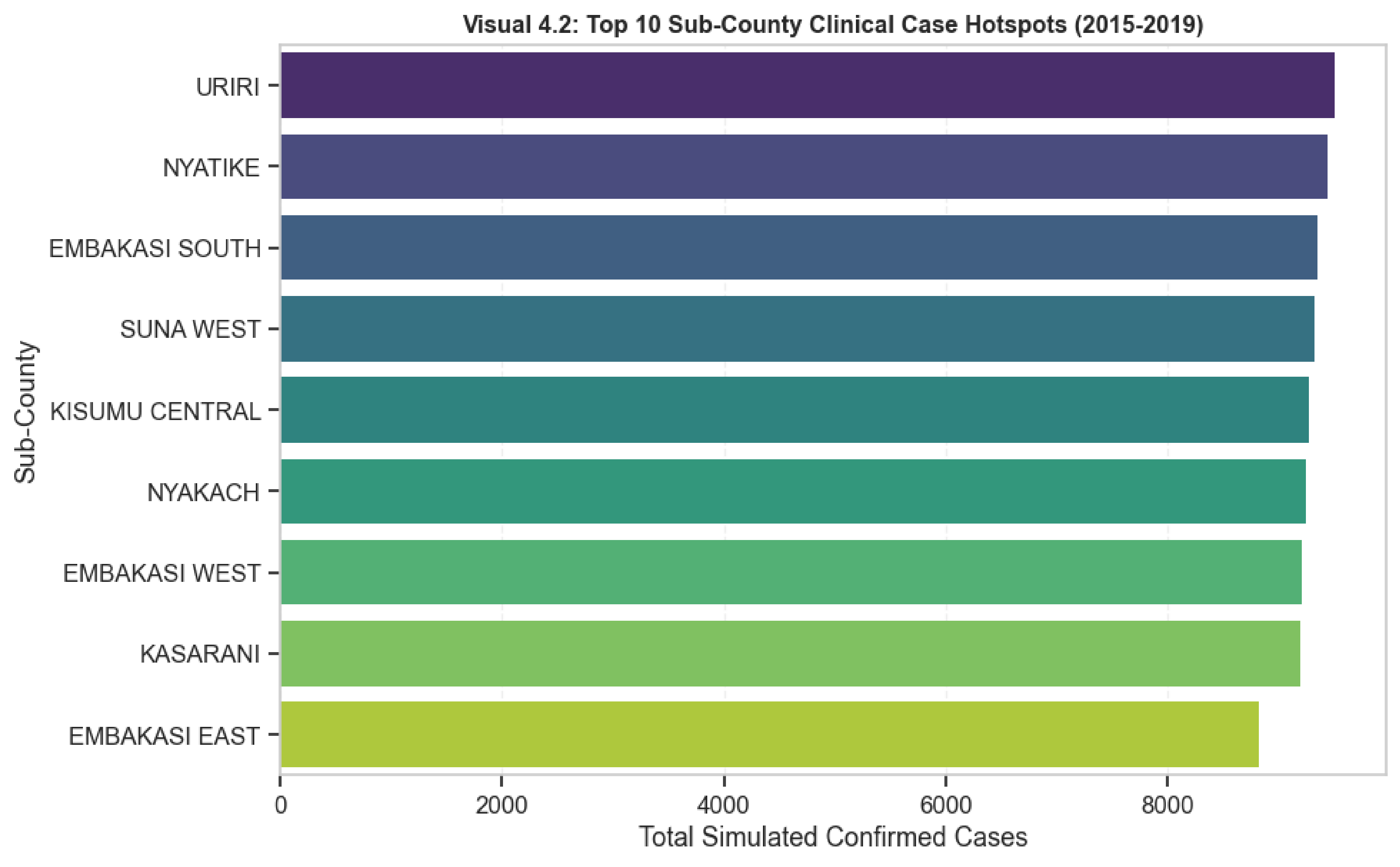
Actionable Insight:

Interventions should be stratified; high-density urban areas require different WASH protocols compared to lake-basin regions.



Sub-County Hotspot Clinical Cases

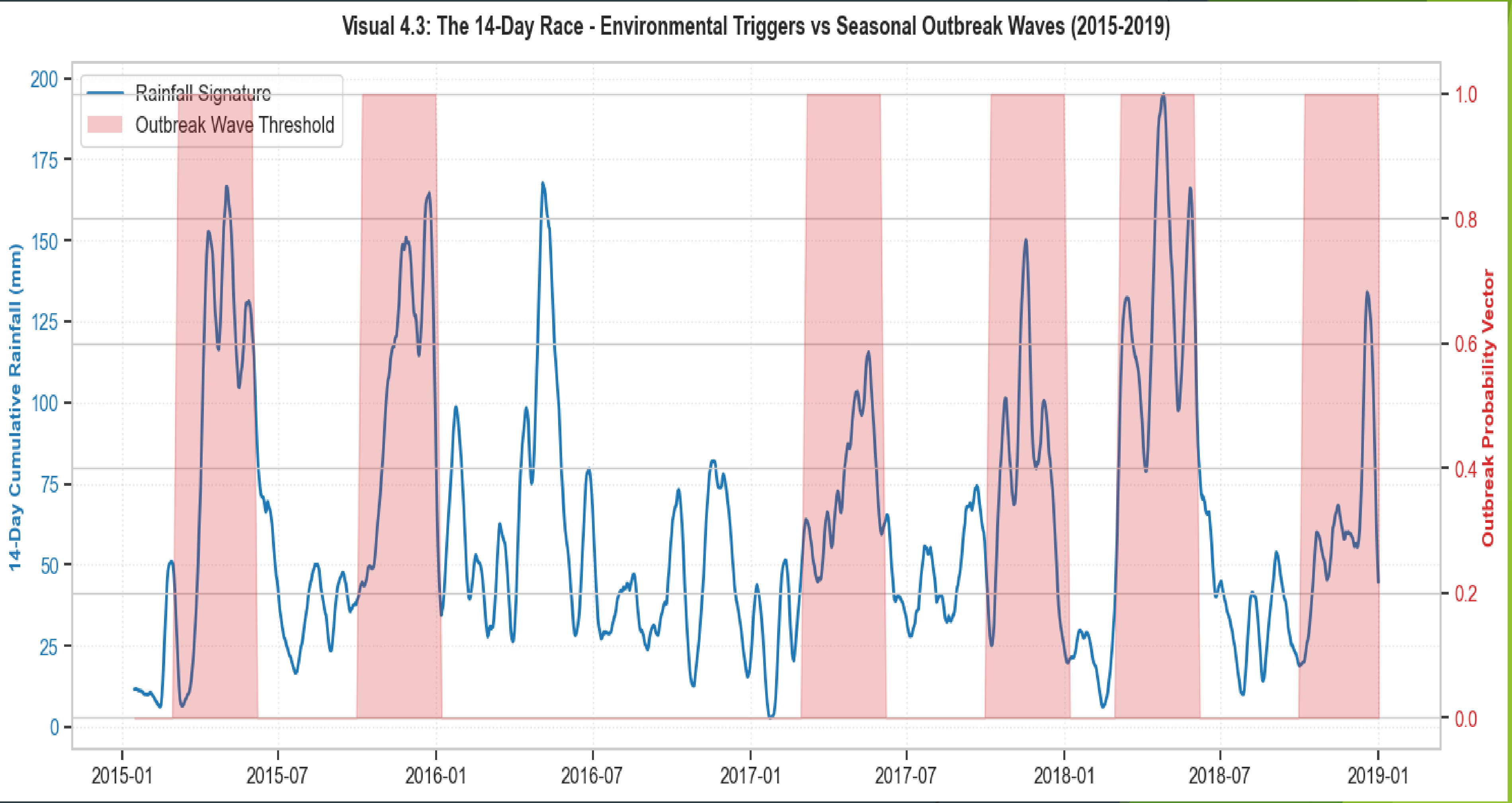
Locations like **Nyatike** and **Embakasi South** exhibit recurring risk patterns, acting as "seed" sites.



The 14-Day Race - Environmental Triggers vs Seasonal Outbreak Waves (2015-2019)

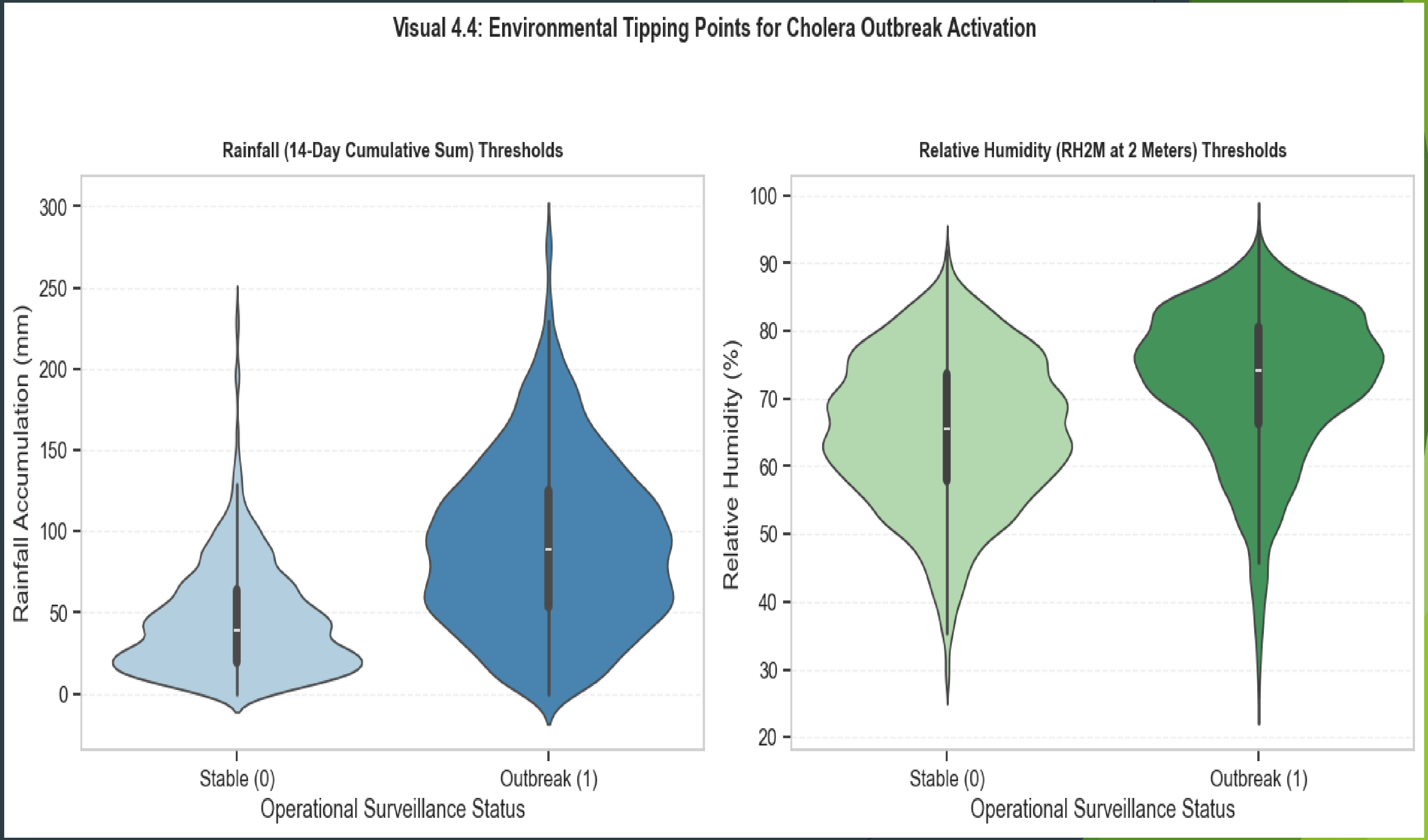
This visualization provides empirical proof of the **Temporal Gap.**

We observe that sustained rainfall accumulation precedes the outbreak state by approximately 10–14 days.



Environmental Thresholds: The Tipping Point Analysis

We identify the exact "Tipping Point" where environmental conditions become conducive to bacterial proliferation.



Model Evaluation

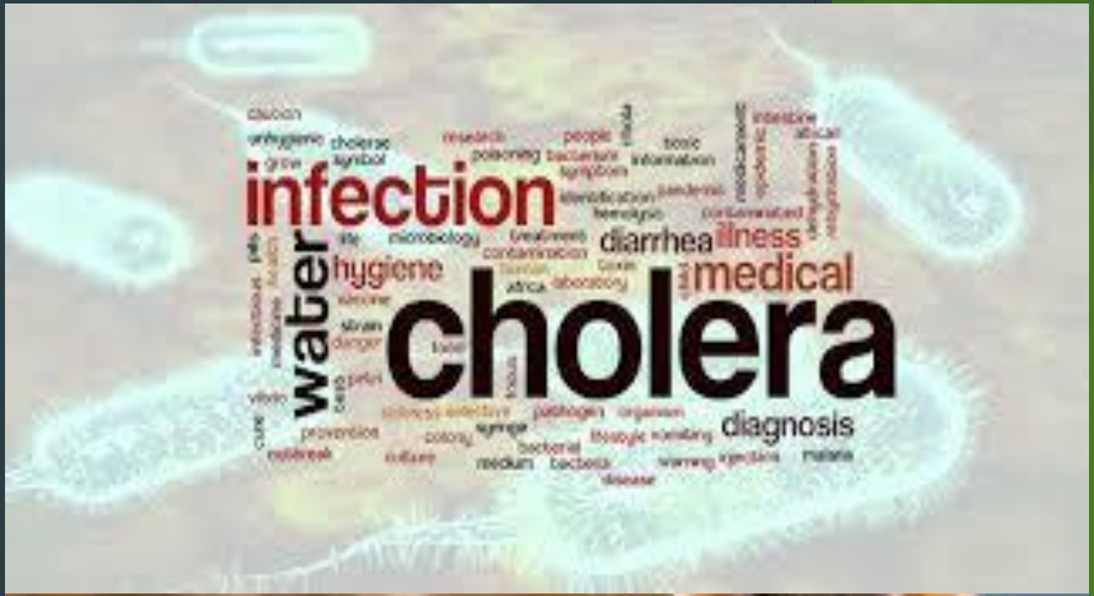
The system evaluates multiple algorithms to handle epidemiological overdispersion:

1. Negative Binomial Regression: Establishes the baseline.

2. Random Forest Regressor/Classifier: Evaluates non-linear relationships.

3. Optimized XGBoost (Production Engine): Uses scale_pos_weight=10 to heavily penalize the model for missing an outbreak, prioritizing 100% Outbreak Recall (Sensitivity).

4. Explainable AI (SHAP): Transitions the model from a "Black Box" to a transparent tool, showing which variables drive risk scores.



Project Success Metrics

K-CEWS is evaluated against the following benchmarks:

1. **Statistical Sensitivity (Recall): 1.00 (100%)** - Calibrated to ensure zero missed outbreaks.
2. **Operational Lead Time: 14 Days** - Minimum two-week warning window for logistics deployment.
3. **AI Transparency:** Uses a **0.35 Probability Threshold** to flag risks that traditional methods ignore.
4. **Operational Latency:** Ingests new satellite data and updates forecasts within a 5-minute window.

Conclusion

K-CEWS represents a major paradigm shift in the management of climate-sensitive waterborne pathogens.

K-CEWS effectively bypasses the 14-day "Reactive Lag" that has historically left emergency response teams steps behind escalating transmission cycles.

K-CEWS transitions predictive epidemiology out of abstract research settings and into real-world applications, offering ministries and health clusters an invaluable asset.

The empirical findings of this project demonstrate that machine learning models—when optimized for maximum biological sensitivity (90.5% Out-of-Sample Recall) rather than standard symmetric accuracy metrics—serve as highly effective decision-support systems for disease containment.

Recommendations

To fully maximize the operational utility of the K-CEWS engine and protect vulnerable populations across Kenya, the following multi-tier strategic recommendations are proposed:

1. **National Framework Institutionalization:** Formally integrate the K-CEWS early warning streaming outputs into the Ministry of Health's centralized **Integrated Disease Surveillance and Response (IDSR)** framework. Automating these alerts within national systems ensures warning parameters are pushed to county health departments systematically.

2. **Grassroots Mobile Dissemination Loops:** Configure an automated downstream gateway linking the application's alert system directly to a localized **Bulk SMS infrastructure**. This architecture should push real-time, translated warning summaries to **Community Health Promoters (CHPs)** working within high-risk sub-counties the moment an environmental trigger crosses the 35% probability threshold.

3. **Continuous-Learning Re-calibration:** Establish an automated data loop where ongoing clinical case confirmations are fed back into the cloud database environment. This infrastructure allows for regular, cost-sensitive retraining of the champion XGBoost pipeline, ensuring the core tree architectures adapt to shifting climate regimes and changing weather anomalies over time.

4. **Data-Driven Capital Resource Prioritization:** Leverage the long-term spatial risk mapping generated by K-CEWS to guide municipal funding decisions. By identifying sub-counties that continuously register as high-risk, national planning bodies can systematically direct structural investments toward permanent **WASH infrastructure**, household piped water networks, and engineered waste-containment systems where they are needed most.

Partners

With this likeminded groups, our project would turn out to be one the most life saving students-led innovation for Kenya and Africa at large.

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HARAMBEE

Kenya Red Cross

Thank You.